

Brucella Epidemiology and Phylogenetic Analysis Using Python

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1. Introduction

Brucellosis is a zoonotic infectious disease caused by bacteria of the genus *Brucella*. Among the most significant species are *Brucella melitensis* and *Brucella abortus*, which are responsible for infections in humans and livestock. The disease is endemic in several regions, including Lebanon, particularly in areas with extensive livestock farming.

The integration of epidemiological data with molecular analysis provides valuable insights into transmission dynamics. This study utilizes Python-based tools to analyze epidemiological data and construct a phylogenetic tree based on DNA sequences, contributing to a One Health approach.

2. Objectives

- Analyze epidemiological data of Brucella infections
- Compare distribution across host species
- Evaluate Ct values as a proxy for bacterial load
- Construct a phylogenetic tree
- Integrate molecular and epidemiological findings

3. Materials and Methods

3.1 Study Design

Retrospective analytical study.

3.2 Dataset

Includes species, host, region, year, Ct value, and sample type.

3.3 Molecular Data

FASTA sequences representing genetic markers (e.g., 16S rRNA, IS711).

Brucella_melitensis_1 ATGCTAGCTAGCTACGATCGATCGATCG

Brucella_abortus_1 ATGCTAGCTAGTTACGATCGATCGATCG

Brucella_melitensis_2 ATGCTAGCTAGCTACGATCGATCGATCA

Brucella_abortus_2 ATGCTAGCTAGTTACGATCGATCGATCA

3.4 Tools

Python, pandas, matplotlib, Biopython.

3.5 Analysis

- Data cleaning
- Calculation of bacterial load ($1/Ct$)
- Visualization (bar, scatter, pie charts)
- Phylogenetic tree construction (Neighbor-Joining method)

4. Results

- Presence of both Brucella melitensis and Brucella abortus

- Distribution across humans and animals

```
== CASES PER HOST ==
host
human      4
goat       4
cow        2
Name: count, dtype: int64
```

- Variation in Ct values

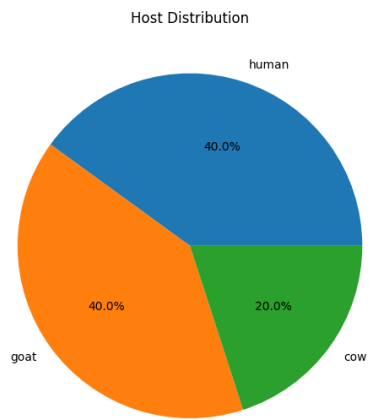
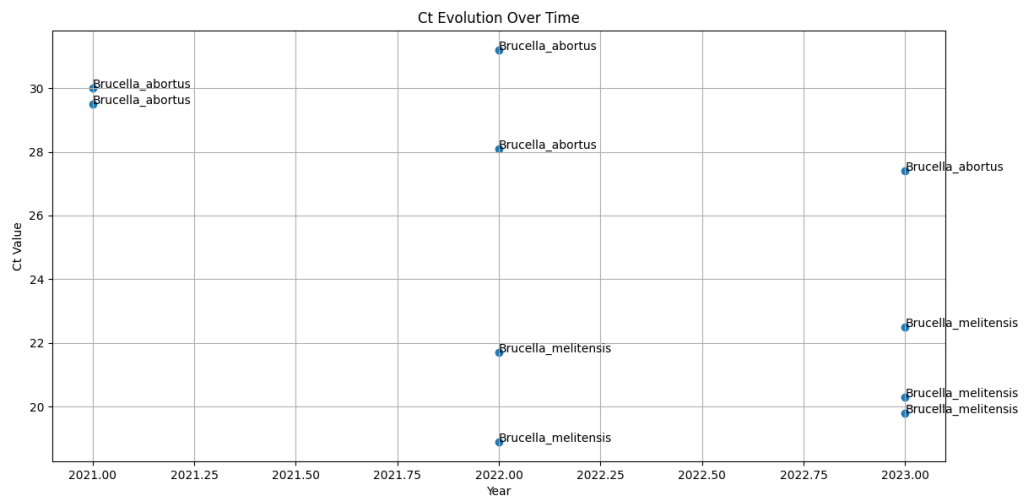
- Phylogenetic clustering by species

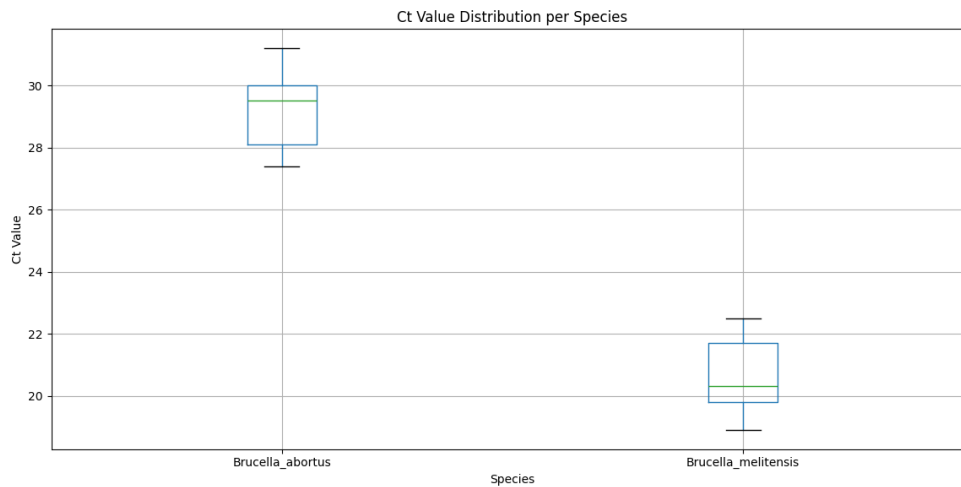
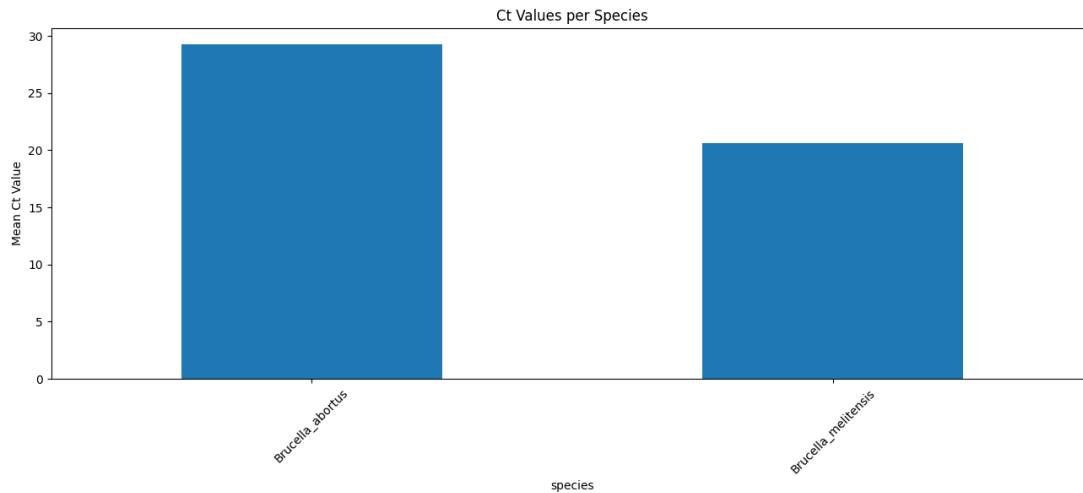
```

=== PHYLOGENETIC TREE (ASCII) ===
                                     Brucella_melitensis_1
      _____|_____
      |                   |
      |                   |_____ Brucella_abortus_1
      |                   |
-  ___|_____ Brucella_melitensis_2
      |
      |_____ Brucella_abortus_2

```

Plot





5. Discussion

The findings highlight the zoonotic transmission of brucellosis and the importance of integrating human and animal data. *Brucella melitensis* appears more associated with human infections. The phylogenetic tree supports genetic relatedness among isolates.

6. Conclusion

This study demonstrates the usefulness of combining epidemiological and molecular data using Python tools to better understand brucellosis transmission within a One Health framework.

